

DEMAND SYSTEMS & INCOME AND SUBSTITUTION EFFECTS—SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

- (a) The optimality condition for an interior solution (i.e., not a corner solution) is

$$\frac{12 + x_2}{10 + x_1} = \frac{p_1}{p_2},$$

Solving the equation above together with the budget line equation $p_1x_1 + p_2x_2 = I$ yields, after some algebra:

$$x_1^m = \frac{I - 10p_1 + 12p_2}{2p_1}, \quad x_2^m = \frac{I + 10p_1 - 12p_2}{2p_2}.$$

Notice that the above quantities are positive if and only if $10p_1 - 12p_2 \leq I$ and $12p_2 - 10p_1 \leq I$. The previous condition is equivalent to $|10p_1 - 12p_2| \leq I$, which is the condition on prices and income that I asked you to assume in order to guarantee an interior solution.

- (b) The quantity demanded of both goods increases with respect to increases in income, therefore, both goods are normal.
- (c) Notice that in both demand expressions, the price of the other good has a positive coefficient. Thus, changes in the price of the other good increase the quantity demanded in both cases. Therefore, the goods are substitutes.
- (d) $x_1^m = \frac{41}{3}$, $x_2^m = 59$.
- (e) $x_1^m = 3$, $x_2^m = 79$.
- (f) With the compensation, the compensated income becomes $I^C = 100 + 34.9086 = 134.9086$. There is no need to solve the expenditure minimization problem to obtain the compensated bundle as well as the required amount of compensation, because we already have the compensated income. We can use I^C and put it into our demand system to obtain the compensated bundle:

$$x_1^C(7, 1, 134.9086) = 5.49347.$$

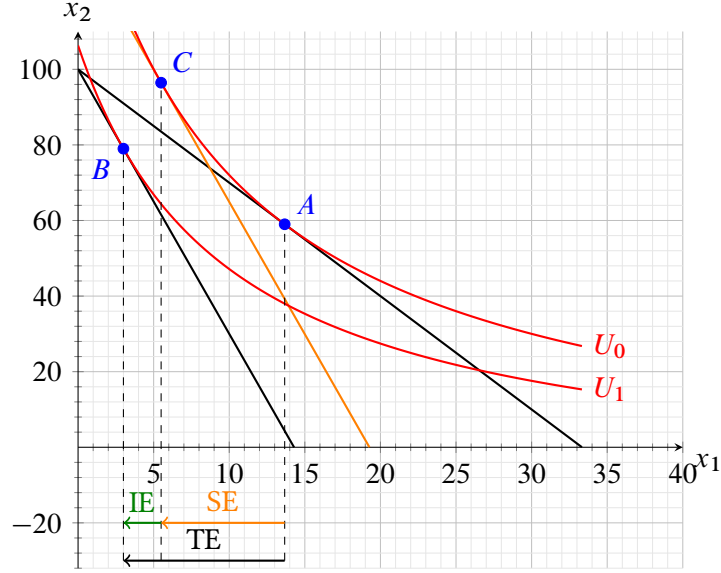
Therefore we have that the substitution, income, and total effects are:

$$SE = 5.49347 - 13.6667 = -8.1732,$$

$$IE = 3 - 5.49347 = -2.49347,$$

$$TE = -10.6667.$$

For an illustration, please see the graph below.



Problem 2

Before solve the items of the problem, let's compute the Marshallian Demand system. The optimality condition is

$$\sqrt{\frac{F}{C}} = \frac{p_C}{p_F},$$

which together with the budget constraint $p_F F + p_C C = I$ identify the solution to the consumer problem. Solving the two equations together yields

$$F^m = \frac{I p_C}{p_F (p_C + p_F)}, \quad C^m = \frac{I p_F}{p_C (p_C + p_F)}.$$

- (a) $F^m = 50, C^m = 50$.
- (b) The price of clothes has increased from 1 to 1.21 due to the ad-valorem tax. The new optimal quantities of food and clothing are: $C^m = 37.3958$, and $F^m = 54.7511$.
- (c) To solve this problem, we need to compute the expenditure minimization problem. Notice that at the original prices, the level of utility achieved was $2\sqrt{50} = 10\sqrt{2}$. We need to compute to what bundle the consumer would substitute when the new price of clothing is $p_C = 1.21$ while attaining the original level of utility $10\sqrt{2}$. Thus the conditions identifying the solution to the expenditure minimization problem are:

$$\sqrt{\frac{F}{C}} = \frac{1.21}{1}, \quad \sqrt{F} + \sqrt{C} = 10\sqrt{2}.$$

Solving the previous system yields the compensated bundles:

$$C^c = 40.9492, \quad F^c = 59.9537.$$

Note that the income required to buy this bundle, is $1.21(40.9402) + 59.9537 = 109.502$. Also notice that if we plug this level of income in the Marshallian demands above we would obtain

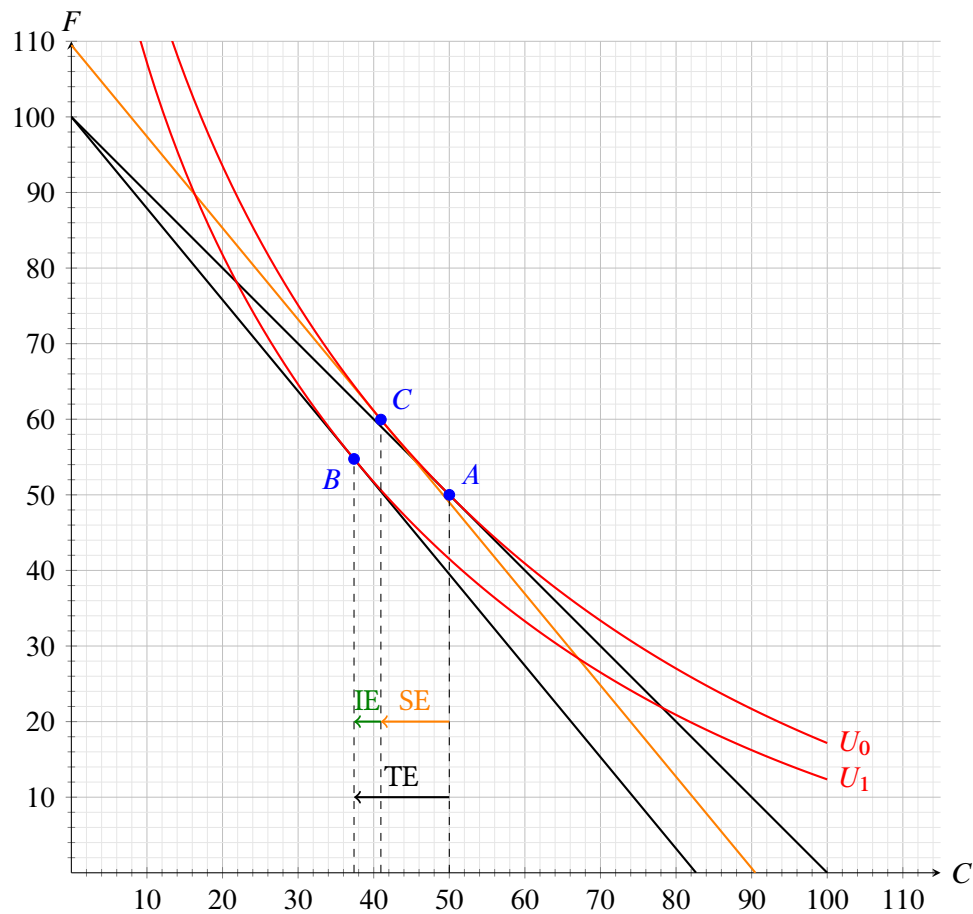
the compensated bundles that we just computed. The substitution, income, and total effects are:

$$SE = 40.9492 - 50 = -9.0508,$$

$$IE = 37.3958 - 40.9492 = -3.55344,$$

$$TE = 37.3958 - 50 = -12.6042.$$

See the picture below for an illustration.



(d) Government revenues are $.21 \times 37.3958 = 7.85311$.

(e) We have computed this previously: since the compensated income required to achieve the original level of utility is 109.502, then the government would need to compensate the individual, i.e., give a cash transfer of $109.502 - 100 = 9.502$ to leave the consumer at the original level of utility.

(f) This is a discussion question. However, here are some important points:

- Consumers react optimally to taxes that introduce distortions in the relative price ratio by altering their consumption patterns.
- In this case, the consumer decided to consume much less of food.
- Government revenues will depend on the final consumption pattern of the consumer.

- If the consumer tries to rebate the tax by giving a cash-transfer back for example, due to the altered consumption pattern of the consumer, government revenues will not be enough to return the consumer to the original level of utility, even with this rebate.
- Thus, in order to avoid this distortion, it is better to impose an *income tax*. Income taxes do not alter the relative price ratio.